A+MENTOR

Maths, Reasoning & English

JKSSB recruitment exam starter notes | Quantitative aptitude • Reasoning • English

How to use these notes

Read the key points, recall the facts without looking, and attempt the practice questions at the end. These are starter notes, not a substitute for the complete syllabus.

1. Arithmetic Essentials

• Percentages: x% of y = (x/100) × y. Percentage change = (change/original) × 100%.

• Profit percentage is usually calculated on cost price; discount percentage is usually calculated on marked price. Read the base carefully.

• Simple interest: SI = (P × R × T)/100. Amount = principal + interest. For compound interest, interest is calculated on the updated balance for each compounding period.

• Ratio a:b compares quantities in the same units. If a:b = m:n, then quantities can be represented as km and kn.

• Average = sum of observations ÷ number of observations. A combined average is a weighted average when group sizes differ.

2. Time, Work, Speed and Number Basics

• If a person completes a job in x days, their one-day work rate is 1/x. Add rates for people working together, assuming independent constant rates.

• Speed = distance/time. Convert units before calculating: 1 km/h = 5/18 m/s; 1 m/s = 18/5 km/h.

• Use divisibility rules, prime factorisation, HCF and LCM for number-system questions. For two positive integers, product = HCF × LCM.

• Linear equation example: 3x + 5 = 20 gives x = 5. Preserve equality by performing the same operation on both sides.

• Mensuration: rectangle area = length × breadth; triangle area = ½ × base × height; circle area = πr^2; circumference = $2\pi r$.

3. Reasoning

• Series: identify the rule—difference, ratio, alternating sequence, squares/cubes or letter positions—before selecting the next term.

• Analogy questions test a consistent relationship, such as function, part-whole, cause-effect or category.

• Direction sense: sketch the route and mark north, east, south and west. Do not rely on mental rotation alone.

• Blood relations: draw a small family tree and label each relationship from the stated person's perspective.

• Syllogisms: judge only what follows logically from the given statements; do not add real-world assumptions.

• Seating arrangements: fix one reference position, record constraints, and eliminate contradictions systematically.

4. English Grammar and Vocabulary

• Parts of speech include nouns, pronouns, verbs, adjectives, adverbs, prepositions, conjunctions and interjections.

• Subject–verb agreement: a singular subject generally takes a singular verb; phrases between subject and verb do not usually change the subject's number.

• Use tense consistently. In reported speech, pronouns, tense and time expressions may change depending on context.

• Common question types include error spotting, sentence improvement, active/passive voice, direct/indirect speech, synonyms, antonyms, idioms, cloze tests and reading comprehension.

• Vocabulary improves with spaced revision. Learn a word with its meaning, part of speech and one example sentence.

5. Exam Practice Method

• First attempt easy questions; mark lengthy questions and return later. Avoid spending too long on one item.

• Use elimination carefully. If there is negative marking, understand the official rules before guessing.

• Maintain an error notebook with topic, mistake, correct method and a similar practice question.

• Always follow the current official notification for the exact JKSSB post, syllabus, language requirements, marks and negative-marking rules.

Practice Questions

1. What is 25% of 240?

A. 40

B. 50

C. 60

D. 80

Answer: C — 60
Explanation: 25% is one quarter; 240 ÷ 4 = 60.

2. A job is completed in 6 days by one worker at a constant rate. The one-day work rate is:

A. 6

B. 1/6

C. 1/3

D. 1/12

Answer: B — 1/6
Explanation: One-day work is the reciprocal of the number of days.

3. A vehicle travels 150 km in 3 hours. Its average speed is:

A. 30 km/h

B. 40 km/h

C. 50 km/h

D. 60 km/h

Answer: C — 50 km/h
Explanation: Speed = distance/time = 150/3 = 50 km/h.

4. The HCF of 12 and 18 is:

A. 2

B. 3

C. 6

D. 36

Answer: C — 6
Explanation: The greatest common divisor of 12 and 18 is 6.

5. Choose the grammatically correct sentence:

A. She go to class daily.

B. She goes to class daily.

C. She going to class daily.

D. She gone to class daily.

Answer: B — She goes to class daily.
Explanation: The singular third-person subject 'she' takes 'goes' in the simple present.

6. If all roses are flowers and some flowers fade quickly, what can be concluded?

A. All roses fade quickly

B. No roses fade quickly

C. All roses are flowers

D. All flowers are roses

Answer: C — All roses are flowers
Explanation: Only the first statement about roses is guaranteed.

7. A person faces north, turns right, then right again. Which direction are they facing?

A. North

B. South

C. East

D. West

Answer: B — South
Explanation: A right turn from north faces east; another right turn faces south.

8. A marked price is ■500 and discount is 10%. Sale price is:

A. ■450

B. ■400

C. ■490

D. ■550

Answer: A — ■450
Explanation: 10% of 500 is 50; 500 – 50 = 450.

Prepared by A+Mentor for learning and revision. Check official notifications and textbooks for the latest syllabus and facts.